

DANIEL JIANG

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EDUCATION

Stanford University, Stanford, CA

Sep 2024 – March 2026

M.S. in Electrical Engineering | Concentration in Software & Hardware Systems

University of Michigan, Ann Arbor, MI

Aug 2020 - May 2024

B.S.E. in Computer Engineering | Minors in Mathematics and Music

summa cum laude

Coursework: Parallel Computing, Operating Systems, Machine Learning, Advanced Topics in Computer Vision, Autonomous Robotics, Robot Perception, Embedded Systems, Data Structures & Algorithms in C++, Linear Dynamical Systems

PROFESSIONAL EXPERIENCE

Waymo, Mountain View, CA

Expected Date: June 2025 – Sep 2025

Incoming Software Engineer Intern | Scalable Mapping Reasoning Team

NVIDIA, Santa Clara, CA

May 2024 – Aug 2024

Software Engineer Intern | Autonomous Driving Ground Truth Perception Team

- Developed a 3D multi-view stereo reconstruction pipeline in C++ and Python using COLMAP, SimpleRecon, and visual-inertial odometry—enabling accurate temporal ground truth generation on LiDAR-less consumer fleet data.
- Integrated and deployed an End-to-End low-latency data factory pipeline using joint ego-motion estimation and camera reconstruction for closed-loop perception ML training & validation.

Aurora, Mountain View, CA

May 2023 – Aug 2023

Software Engineer Intern | Autonomous Driving Onboard Localization Team

- Built a C++ lane-detection verification system to identify erroneous EKF inputs and support lane-level alignment through geometric analysis with Bezier interpolation; deployed on all vehicles with 97% precision and 80% recall.
- Designed a low-latency LiDAR instance segmentation C++ library using a novel momentum-based DBSCAN point cloud clustering algorithm; deployed onboard for 3D lane detection segmentation.

Robert Bosch LLC, Plymouth, MI

May 2022 – Aug 2022

Software Engineer Intern | Advanced Driver Assistance Systems Team

- Adapted an algorithm to detect co-channel FMCW radar interference and characterize sections of detection data by radar velocity, validating the interference pattern of Bosch's parking radar sensors.

RESEARCH EXPERIENCE

Stanford Robotics and Embodied Artificial Intelligence Lab (REAL), Stanford, CA

Dec 2024 - Present

Research Assistant | Advisors: Prof. Shuran Song & Prof. Karen Liu

- Developing a monocular depth estimation pipeline using Depth Anything V2 and onboard low-latency visual-inertial odometry capability for End-to-End humanoid loco-manipulation imitation learning.

University of Michigan Transportation Research Institute, Ann Arbor, MI

May 2021 – May 2024

Undergraduate Research Assistant | Advisor: Prof. Matthew Reed

- Published open-source GUI software using Python, Qt, and VTK to generate 3D parametric finite-element body shape models for crash protection studies; awarded 2021 Michigan Student Research Competition 3rd place honor.
- Devised a specialized ICP alignment program and a watertight surface reconstruction method for Kinect 3D scans.
- Released a Python library for visualizing and manipulating LS-DYNA human mesh data for ergonomics analysis.

Baraja (Australian LiDAR Startup) with TechLab at Mcity, Ann Arbor, MI

Aug 2022 – April 2023

LiDAR Research Cohort Member

- Built an automated Python verification system to evaluate repeatability, probability of detection, and coherent Doppler velocity of Baraja's solid-state RMCW LiDAR sensors in adverse meteorological conditions.

LEADERSHIP EXPERIENCE

Michigan Mars Rover Team, Ann Arbor, MI

October 2020 – May 2024

Simulation Team Lead & Senior Technical Advisor

- Placed 1st in both the University Rover Challenge (out of 99 universities) and the Canadian International Rover Challenge in 2022; led a team of 20 engineers in developing the Navigation Simulation System (demo).
- Developed an Unreal Engine C++ 3D simulator and a TypeScript trajectory simulator to replicate rover's behavior on Mars' terrain, simulate IMU Gaussian noise, and enable remote algorithm testing during the pandemic.

SKILLS

Programming: C++, Python, Bash, C, Verilog, JavaScript, OpenCV, OpenGL, PyTorch, Qt, NumPy, Eigen, Protobuf

Technologies: LiDAR, IMU, Radar, Linux, VS Code, Unreal Engine, MATLAB, ROS, Gazebo, Quartus, Bazel, Git

Interests: Automobiles, Basketball Analytics, Skiing, Poker, Campus Symphony Orchestra, Venetian Seafood Risotto